GENERATIVE AI / RESEARCH PROJECT

Occlusion-Aware Frame Interpolation

Learning which warped frame to trust at each pixel

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Problem: reconstruct the missing middle frame

Input: two endpoint frames. Output: the frame halfway between them.

$$(I_0, I_1) \longrightarrow \hat{I}_{1/2}$$

Why is it hard?

Motion shifts pixels; occlusion makes some content visible in only one endpoint. Averaging imperfect warps can produce double edges and ghosting.

Our solution

Freeze RAFT-small and share the same warps across all methods. Train a 95k-parameter CNN to choose the blending weight at each pixel.

Research question

Does a learned mask improve reconstruction over fixed and flow-consistency blending, especially near motion boundaries and occlusion-related regions?

The middle frame is ground truth during training/evaluation only. A mask cannot invent detail missing from both warps.

Method, experimental setups and data

All primary methods use identical frozen flow estimates and midpoint warps.

1 Estimate motion and warp to the midpoint

$$F_{ij} = \mathrm{RAFT}(I_i, I_j), \quad W_i = \mathcal{S}(I_i, \tfrac{1}{2}F_{ij})$$

i, j ∈ {0,1}, j ≠ i. S: weighted forward splat + hole fallback.

2 Compare three blending rules

$$\hat{I}_{1/2} = M \odot W_0 + (1 - M) \odot W_1$$

Fixed: $M = 0.5$

Heuristic: $M = \mathrm{clip}_{[0,1]}[0.5 + 0.5s(O_0 - O_1)]$

Ours: $M = \sigma(g_\theta(X))$ · **18-channel input**

O: midpoint flow-consistency proxy; explicit hole overrides.
X: warps, differences, flow, visibility/coverage and heuristic prior.

3 Train the mask with weighted reconstruction loss

$$\mathcal{L} = \frac{\sum_p w_p \|I_p - I_p^{GT}\|_1}{3\sum_p w_p}, \quad w_p = 1 + 4D_p$$

D = 1 where mean RGB warp disagreement > 0.04.

Data & framework

Vimeo-90K triplets
Train / val / test:
1,200 / 150 / 3,782
Resolution: 256 × 448

PyTorch + torchvision
Frozen RAFT-small · H100
Mask: 8k steps, 3 seeds
Heuristic s = 0.5 (validation)

Additional L1 / FM refiners
L1: 12k; FM: 12k / 30k; EMA
Euler / Heun: 20 / 50 / 100
Logit-normal; CFG 1 / 2 / 3; mean of 4

Pinned mirrors; exact frame overlaps excluded.
Similar scenes may remain. Refiner tuning: 64 val triplets.

Results: the learned mask improves reconstruction

Full test: 3,782 triplets. ↑ higher is better; ↓ lower is better. Green marks improvements.

Primary comparison	PSNR ↑	SSIM ↑	MAE ↓	PSNR occ* ↑
Baseline 1 · fixed	32.583	0.94730	0.011315	24.138
Baseline 2 · heuristic	32.664	0.94769	0.011159	24.170
OURS · learned mask	**32.917**	**0.95076**	**0.010812**	**24.617**
Gain vs fixed	**+0.334**	**+0.00347**	**-0.000503**	**+0.479**

ΔPSNR 95% CI [+0.314, +0.356] dB · all 3 seeds improve · without 16px border: +0.234 dB

Separate extension (3.31M params)	PSNR ↑	Δ vs mask	LPIPS ↓
Mask only (reference)	32.917	—	0.0413
+ L1 refiner	**33.405**	**+0.489**	**0.0342**
+ FM refiner	32.031	-0.886	0.0513
+ FM, mean of 4	32.704	-0.213	0.0374

L1 wins both metrics.
FM averaging improves LPIPS,
but loses PSNR vs mask-only.

*occ is a flow/coverage proxy, not a visibility label; regional PSNR pools error. Other PSNR values average images. CI resamples triplets.

Demo: baseline versus our learned mask

Three held-out examples: improvement, typical behavior and a failure case.

BASELINE
Fixed 50/50 blending

OUR METHOD
Learned blending mask

REFERENCE
Ground-truth middle frame



00022/0351 | interpolated midpoint | three-frame loop, slowed for inspection

Click the video in PowerPoint slide-show mode to play.

Slowed three-frame sequences: input endpoint → predicted / true midpoint → input endpoint.
The MP4 is embedded in the PPTX. This PDF displays a still frame.

Related work and our contribution

Learned visibility-aware blending is an established idea.

Prior work	Established idea	Our experimental scope
Super SloMo · Jiang et al. CVPR 2018	Visibility maps + flow refinement	Frozen flow; only the mask is trained
Softmax Splatting · Niklaus & Liu CVPR 2020	Weighted forward warping + image synthesis	Shared splats across all three methods
RIFE · Huang et al. ECCV 2022	Directly learned intermediate flow	We isolate the effect of blending

Our contribution: a controlled empirical comparison

Frozen RAFT-small · lightweight mask · corrected splits · uncertainty · failure cases

Not a new interpolation algorithm, not an exact reproduction of a prior paper, and not DLSS.

Backbone: RAFT (Teed & Deng, 2020). Extension: Flow Matching (Lipman et al., 2022). Paper links in PPTX notes and notebook.